

Felipe Felix Arias

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SUMMARY

MLE working across ML training infrastructure, GenAI systems, and evaluation pipelines. Experienced spanning the full ML stack: distributed GPU training, model fine-tuning, agentic applications, and structured evaluation. Research background in programmatic data labeling and weak supervision (NSF fellow, published with Google Brain at IEEE ICRA).

SKILLS

Languages: Python, PyTorch, Go, SQL
Infra: Kubernetes, Docker, AWS, Spark

ML: DDP, FSDP, DeepSpeed, Ray, LoRA/QLoRA, XGBoost
GenAI: Agentic systems, RAG, eval pipelines, Claude Code, Codex

EXPERIENCE

Uber Technologies – ML Training Team

San Francisco, CA

Machine Learning Engineer

Mar 2024 – Present

- Led 8-month migration of Uber’s core ML training scheduler from zonal to regional federated compute, migrating 2,000+ pipelines with zero downtime. Co-designed Ray-on-Kubernetes orchestration with custom controllers for cross-cluster GPU scheduling of PyTorch/DeepSpeed workloads.
- Built multi-signal safety system: GenAI transcript analysis proposes distracted driving flags, classification NNs on telemetry validate, achieving a 3.7% reduction in unsafe behaviors. Designed generation/decision separation pattern for safe LLM-in-the-loop production systems.
- Built and deployed RAG-based incident assistant (company-wide) and employee information agent (~130 MAU). Point of contact for LLM fine-tuning partnerships (Fireworks AI), advising 12+ teams on fine-tuning workflows and evaluation.

Marovi AI

San Francisco, CA

Founder

2024 – Present

- Building research comprehension and translation platform with provider-agnostic LLM API (OpenAI, Anthropic, Google, DeepL), agentic translation pipelines with glossary enforcement and structured output, and crowdsourced correction loops across 9+ languages. Top 10% at Y Combinator twice.
- Built evaluation infrastructure: benchmark frameworks (FLORES, XL-SUM), LLM-as-judge systems for automated quality scoring, and A/B comparison across providers. Full-stack: AWS, Python, Pydantic schemas, caching, and batching.

University of Illinois at Urbana-Champaign – Parasol Lab

Urbana, IL

NSF Graduate Research Fellow – Advisor: Nancy M. Amato

2019 – 2023

- Designed self-supervised programmatic data labeling pipelines: simulation generates collision data at scale, spatial regression models distill learned structure into planning heuristics, achieving 5x speedup with zero manual labels. Published with Google Brain researchers at IEEE ICRA 2021.
- Released open-source simulation and evaluation tools for multi-agent datasets. Master’s thesis on motion pattern prediction using learned spatial representations.

Uber Technologies – Search Team

San Francisco, CA

ML Engineering Intern

Jun – Aug 2023

- Developed XGBoost and DNN models for ETA prediction in driver-rider matching. Built dual-metric evaluation framework (regression accuracy + Spearman rank correlation) for low-latency inference and ranking quality.

Stanford University – Hazy Research Lab

Stanford, CA

Research – Advisor: Christopher Ré

2018 – 2019

- Extended Snorkel’s programmatic data labeling for multi-sentence relation extraction with Alex Ratner (now CEO of Snorkel AI). Designed novel labeling functions and multi-task learning strategies, achieving 12% F1 improvement via LSTMs.

EDUCATION

University of Illinois at Urbana-Champaign

2023

M.S. Computer Science (GPA: 4.0/4.0) – Fellowships: NSF GRF, Ford HM, GEM

University of Illinois at Urbana-Champaign

2019

B.S. Computer Science with Honors (GPA: 3.8/4.0)

PUBLICATIONS

Motion Pattern Prediction in Dynamic Environments (Master’s Thesis) – UIUC, 2023

Avoidance Critical Probabilistic Roadmaps for Motion Planning in Dynamic Environments

F. F. Arias, B. Ichter, A. Faust, N. M. Amato – IEEE ICRA 2021